Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	21/10/2026
Team ID	Suhani Porwal
Project Name	**Electric Vehicle Market Performance & Insights**
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)	Priority (High/Medium/Low)
1	Data Access & Integration	The solution shall connect to the EV market datasets containing information such as vehicle category, manufacturer, annual sales, battery technology, battery capacity, driving range, manufacturing country, and customer ratings. The data shall be imported into Tableau for analysis and visualization.	High
2	Data Processing	The solution shall prepare the EV datasets by checking data quality, handling missing or inconsistent values, organizing fields into appropriate dimensions and measures, and applying Tableau filters and calculated fields where required for analysis.	High
3	Business Analysis	The solution shall calculate and analyze key metrics such as total annual sales, average driving range, average battery capacity, average customer rating, manufacturer performance, vehicle category performance, battery technology distribution, and manufacturing country distribution.	High
4	Business Insights & Analytical Outputs	The solution shall provide interactive Tableau dashboards containing KPIs, charts, comparisons, trends, and analytical views to identify leading manufacturers, high-performing vehicle categories, dominant battery technologies, manufacturing trends, customer satisfaction, and other important EV market insights.	High

5	User Interaction & Analysis	Users shall be able to interact with the dashboard using filters and selections based on relevant fields such as manufacturer, vehicle category, battery technology, manufacturing country, and other available attributes. This shall allow users to compare different segments and explore specific market trends.	High
6	Data Update & Reporting	The solution shall allow the Tableau workbook to be refreshed when the underlying EV datasets is updated. The dashboards shall present the latest available data through updated KPIs, visualizations, and analytical outputs for reporting and decision-making.	Medium

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	The Tableau dashboard shall have a clear, simple, and user-friendly layout. KPIs, charts, filters, and other visual elements shall be arranged logically so that users can easily understand EV market performance and obtain insights without extensive technical knowledge.
NFR-2	**Security**	The project shall use only the required EV market data and shall avoid exposing confidential or personally identifiable information. Access to the Tableau workbook and dashboard shall be controlled through the applicable Tableau sharing and access settings.
NFR-3	**Reliability**	The dashboard shall provide consistent and accurate results based on the underlying datasets and defined calculations. Charts, KPIs, filters, and calculated fields shall function correctly and produce reliable analytical outputs.
NFR-4	**Performance**	The Tableau dashboard shall load and respond efficiently to user interactions such as filtering, selecting categories, and comparing manufacturers. Visualizations shall be designed to avoid unnecessary complexity and maintain smooth dashboard performance.
NFR-5	**Availability**	The completed Tableau dashboard shall remain accessible to the intended users whenever the Tableau workbook or published dashboard is available. The solution shall provide continuous access to the available EV market analysis without requiring separate analytical software or processes.
NFR-6	**Scalability**	The Tableau solution shall be designed so that additional EV records, manufacturers, vehicle categories, battery technologies, or geographical data can be incorporated into the datasets and analyzed using the existing dashboard structure with minimal modification.